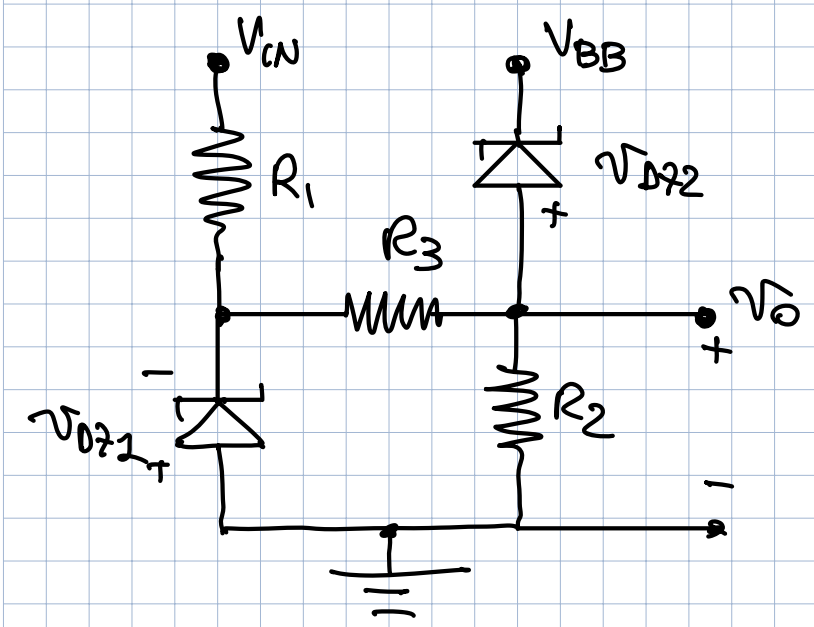
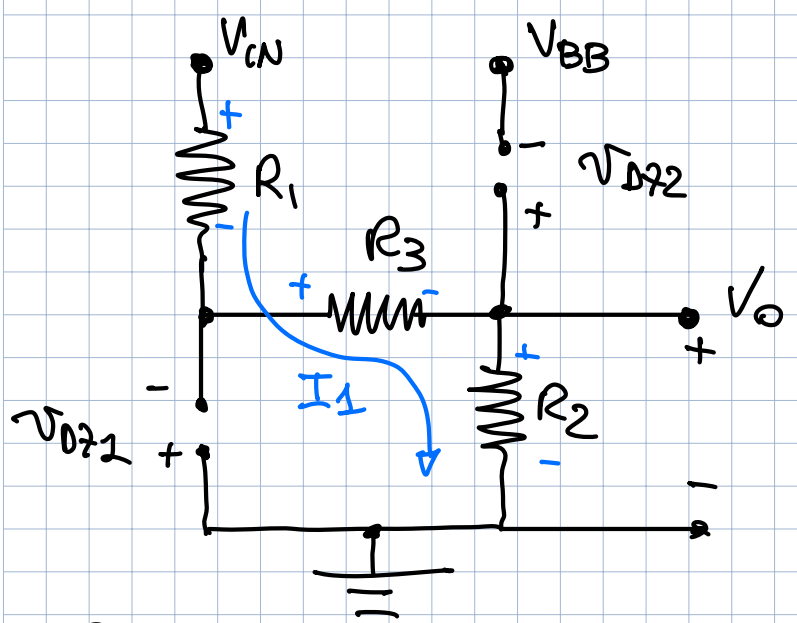
EDP2 - 5



$R_1 = 2K\Omega$
$R_2 = 3K\Omega$
$R_3 = 10\ K\Omega$
$V_{BB} = 18V$
$V_{ON} = 0V$
$V_{Z1} = 5V$
$V_{Z2} = 8V$

TROVARE v_O QUANDO $V_{IN} = -20V$, $V_{IN} = 10V$, $V_{IN} = 30V$

HP DZ_1 e DZ_2 = OFF



$$I_1 = \frac{V_{IN}}{R_1+R_2+R_3}$$

$$v_{DZ1} = -V_{IN}\frac{R_2+R_3}{R_1+R_2+R_3} = -V_{IN}\cdot\frac{13}{15}$$

$$v_{DZ2} = +V_{IN}\frac{R_2}{R_1+R_2+R_3} - V_{BB} = V_{IN}\cdot\frac{3}{15} - V_{BB}$$

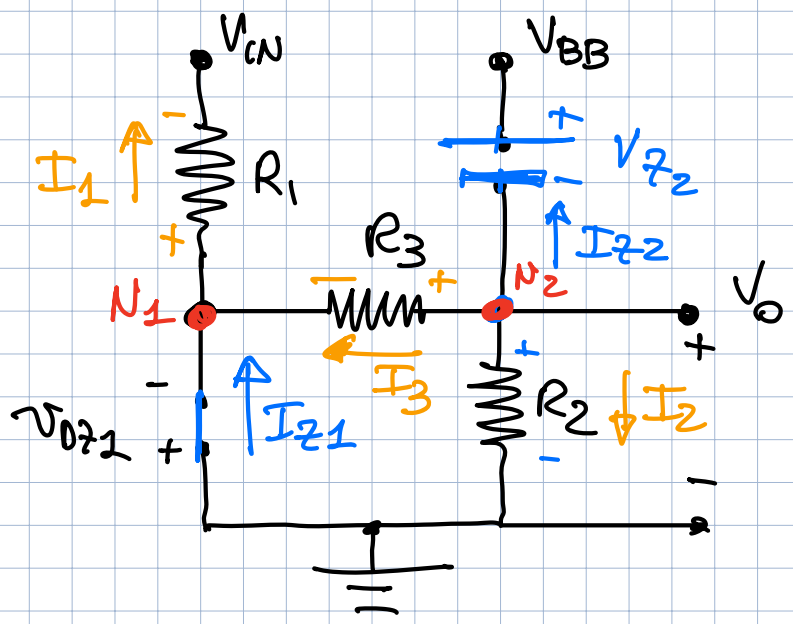
SE

$V_{IN} = -20V$ $\quad v_{DZ1} = 17{,}3\overline{3}\ V > 0$

$v_{DZ2} = -22V < -V_{Z2} \Rightarrow$ BREAK

HP $DZ1$ = ON
$DZ2$ = BREAK

$\Downarrow$



$$V_o = V_{BB} - V_{Z2} = 10V$$

$$I_2 = \frac{V_o}{R_2} = \frac{10V}{3K\Omega} = 3,3\overline{3}\,mA$$

$$I_3 = \frac{V_o}{R_3} = \frac{10V}{10K\Omega} = 1\,mA$$

N_2

$$\Rightarrow I_{Z2} = -I_3 - I_2 = -4,3\overline{3}\,mA$$

<0 OK BR.

$$I_1 = -\frac{V_{IN}}{R_1} = \frac{20V}{2K\Omega} = 10\,mA$$

N_1: $I_1 = I_{Z1} + I_3 \Rightarrow I_{Z1} = I_1 - I_3 = 9\,mA > 0$

OK $D1 = ON$

QUINDI CON $V_{IN} = -20V$

$D_{Z1} = ON$
$D_{Z2} = BREAK.$ $\Rightarrow V_o = 10V$

SE $V_{IN} = 10V$

$$V_{DZ1} = -V_{IN} \cdot \frac{13}{15} = -8,67V < V_{Z1}$$

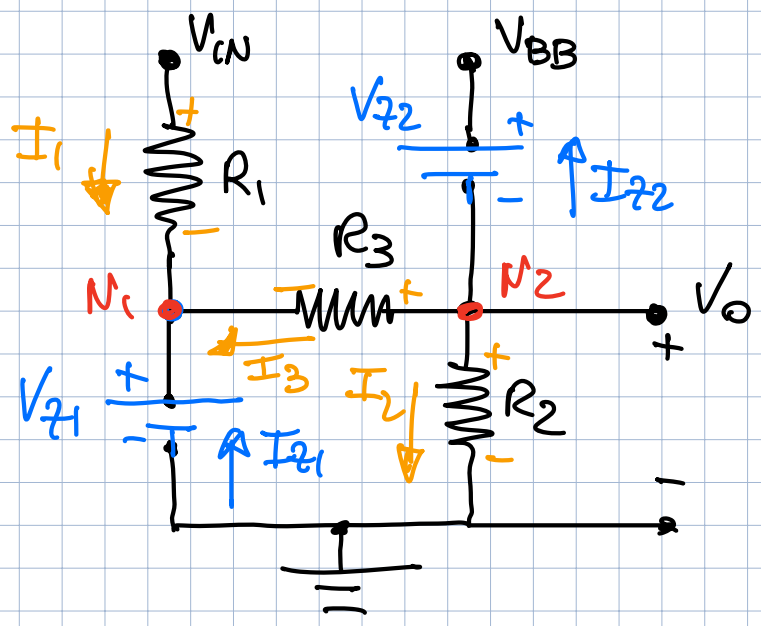
$$V_{DZ2} = V_{IN} \cdot \frac{3}{15} - V_{BB} = 2V - 18V = -16V < V_{Z2}$$

HP D_{Z1} e D_{Z2} = BREAK

$$I_1 = \frac{V_{IN} - V_{Z1}}{R_1} = \frac{10V - 5V}{2K\Omega} = 2,5\,mA$$

$$V_o = V_{BB} - V_{Z2} = 18V - 8V = 10V$$

$$I_2 = \frac{V_o}{R_2} = \frac{10V}{3K\Omega} = 3,3\overline{3}\,mA$$



$$I_3 = \frac{V_O - V_{Z1}}{R_3} = \frac{5V}{10K\Omega} = 0{,}5\ mA$$

N_1: $I_{Z1} = -I_1 - I_3 = -2\ mA < 0$

OK BREAK

$$I_{Z2} = -I_2 - I_3 = -3{,}83\ mA < 0$$

OK BREAK

SE $V_{IN} = 30V$

IL CIRCUITO RIMANE COME X $V_{IN} = 10V$

L'UNICA DIFFERENZA È CHE

$$I_1 = \frac{V_{IN} - V_{Z1}}{R_1} = \frac{25V}{2K\Omega} = 12{,}5\ mA$$

$$I_{Z1} = -I_1 - I_3 = -13\ mA$$

PIÙ INTERESSANTE ERA CONSIDERARE $V_{IN} = 5V$

HP $D_{Z1} = D_{Z2} = OFF$ (VEDI SOPRA)

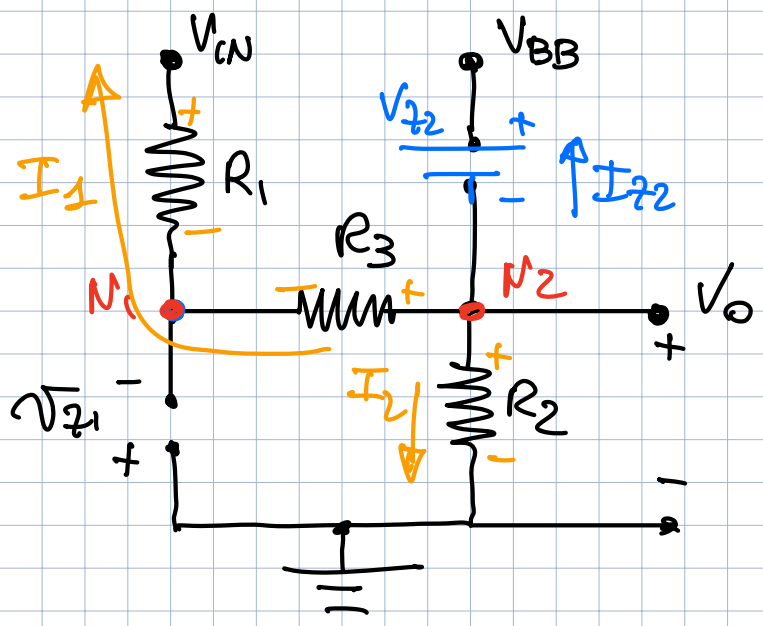
$$V_{DZ1} = -V_{IN} \cdot \frac{13}{15} = -4{,}33V < V_{Z1}$$

$$V_{DZ2} = V_{IN} \cdot \frac{3}{15} - V_B = 1V - 18V = -17V < V_{Z2}$$

HP $D_{Z1} = OFF$

$D_{Z2} = BREAK$

$$V_O = V_{BB} - V_{Z2} = 10V$$



$$I_2 = \frac{V_O}{R_2} = 3{,}33\,mA$$

$$I_1 = \frac{V_O - V_{IN}}{R_1 + R_3} = \frac{5V}{12K\Omega} = 0{,}417\,mA$$

$$I_{Z2} = -I_1 - I_2 = -3{,}75\,mA < 0$$

OK BREAK

$$V_{N_1} = V_O - R_3 I_1 = 10V - 10K\Omega \cdot 0{,}417\,mA = 5{,}83V$$

$$V_{DZ1} = -V_{N_1} = -5{,}83V < V_{Z1}$$

$\Rightarrow$ DZ1 È IN BREAKDOWN

$\Rightarrow$ VEDI CASO $V_{IN} = 10V$ (ORA CON $V_{IN} = 5V$)

$$I_1 = \frac{V_{IN} - V_Z}{R_1} = 0V$$

$V_O = 10V$

$I_2 = 3{,}33\,mA$

$I_3 = 0{,}5\,mA$

$$I_{Z1} = -I_1 - I_3 = -I_3 = -0{,}5\,mA < 0$$

OK Z1 = BREAK

$$I_{Z2} = -I_2 - I_3 = -3{,}83\,mA < 0$$

OK Z2 BREAK

ORA CONSIDERIAMO $V_{IN} = -5V$

HP $D_{Z1} = D_{Z2} = OFF$ (VEDI SOPRA)

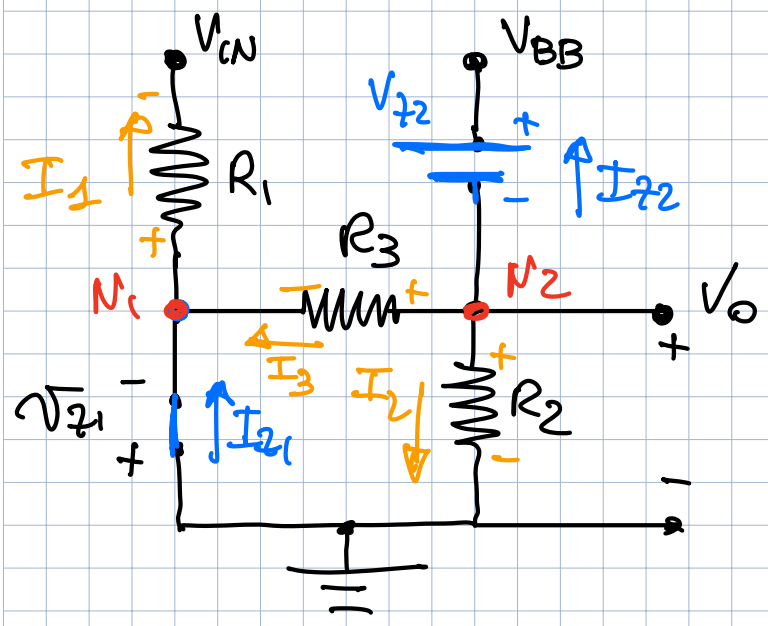
$$V_{DZ1} = -V_{IN} \cdot \frac{13}{15} = +4{,}33\,V > 0 \qquad D_{Z1} = ON\,?$$

$$V_{DZ2} = V_{IN} \cdot \frac{3}{15} - V_B = -1 - 18V = -19\,V < V_{Z2}$$

HP $D_{Z1} = ON$

$D_{Z2} = BREAK$

$$V_O = V_{BB} - V_{Z2} = 10V$$



$$I_2 = \frac{V_O}{R_2} = 3{,}33\,mA$$

$$I_3 = \frac{V_O}{R_3} = \frac{10V}{10K\Omega} = 1\,mA$$

$$I_{Z2} = -I_2 - I_3 = -4{,}33 < 0$$

OK DZ2 BREAK

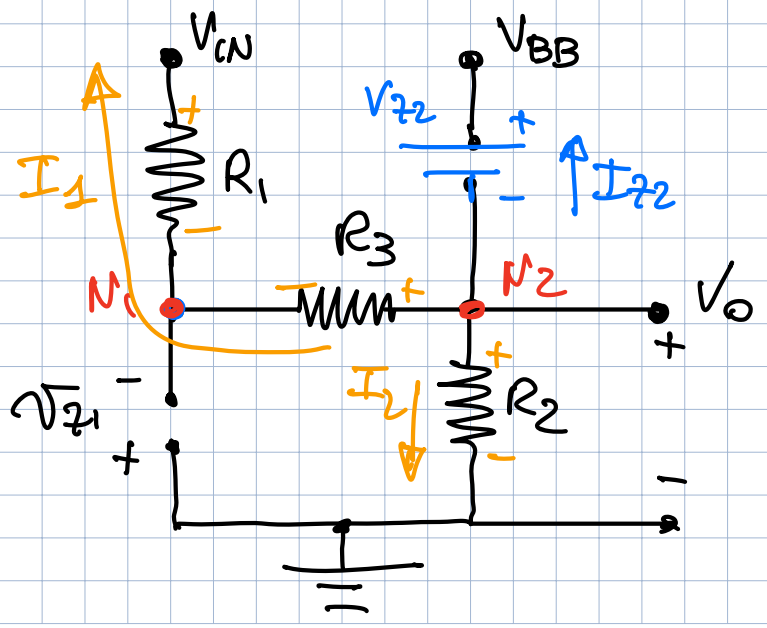
$$I_1 = \frac{-V_{IN}}{R_1} = -2{,}5\,mA$$

$$I_{Z1} = I_1 - I_3 = -1{,}5\,mA < 0$$

$\Rightarrow$ DZ1 NON È ON MA OFF

$\Rightarrow$ $V_{IN} = -5V$

$D_1 = OFF$, $D_2 = BREAK$



$$V_O = V_{BB} - V_{Z2} = 10V$$

$$I_2 = \frac{V_O}{R_2} = 3{,}33\,mA$$

$$I_1 = \frac{V_O - V_{IN}}{R_1 + R_3} = \frac{10V}{12K\Omega} = 0{,}833\,mA$$

$I_{Z2} = -I_1 - I_2 = -4{,}1\bar{6}\ mA < 0$ OK BREAK

$V_{N1} = V_0 - R_3 I_1 = 10V - 10K\Omega \cdot 0{,}83\bar{3}\ mA$

$= 1{,}67\ V$

$V_{DZ1} = -V_{N1} = -1{,}67\ V > -V_Z \Rightarrow$ DZ1 è OFF